

SFM-Adapter: Style-Aware Feature Manipulation Adapter for Speech Style Editing

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Abstract—Speech Style Editing (SSE) aims to modify selected style attributes (e.g., timbre, emotion, pitch) while preserving the linguistic content and all other style attributes that are not given. Many speech applications require flexible control over speech style, making SSE increasingly important. Existing SSE approaches typically follow a style-generation paradigm that synthesizes non-linguistic attributes from style conditions. However, this often results in limited preservation of source attributes and insufficient flexibility when only a subset of style attributes is specified. To overcome these limitations, we adopt a style editing paradigm, in which the target style is achieved by adjusting the source speech instead of producing speech from scratch. Building on this paradigm, we propose a diffusion-based framework with a Style-aware Feature Manipulation Adapter (SFM-Adapter). The SFM-Adapter performs feature-level modulation by integrating user-provided style information with source speech features through multi-layer cross-attention. The resulting modulated features are incorporated into the generation process via mask attention. During inference, a Large Audio-Language Model (LALM)-based length regulation is designed to predict speaking speed and adjust duration. Experiments across multiple speech style editing tasks demonstrate that the SFM-Adapter achieves more natural, accurate, and source-preserving style editing compared with existing methods.

Index Terms—Diffusion models, feature manipulation, speech style editing, style-aware.

I. INTRODUCTION

SPEECH style plays an important role in human communication, as it can convey a speaker’s emotion, attitude, and even mental state [1]. In speech style modeling, style-related attributes are typically described using two major groups: (1) prosodic or expressive features, such as pitch, duration, and speaking rate, and (2) timbre or voice-quality features that capture speaker-dependent spectral characteristics [1], [2]. While timbre contributes to the identity and texture of the voice, expressive features shape how the message is delivered, making them both essential in speech style modeling. Understanding and modeling speech style holds significant value for a wide range of applications, including personalised speech synthesis [3], voice dubbing [4], and voice cloning [5]. Many of these applications require the ability to control or manipulate specific style attributes of speech, which has led to increasing interest in speech style editing (SSE) [3], [6].

In the past, voice style transfer and voice conversion methods were commonly used as speech style editing approaches, achieving progress in timbre editing [3], [7], [8] and expressiveness editing [9], [10], [11], [12]. However, these methods handle style dimensions such as emotion, timbre, speaking rate, and pitch independently, preventing them from modeling their interactions. For example, these may convert “sadness” to “excitement” without modeling how different style attributes interact. In practice, “sadness” is typically associated with lower pitch, whereas “excitement” often involves rising pitch. Moreover, pitch distributions vary significantly across different timbres (e.g., male vs. female or child vs. adult speakers), even when expressing the same emotion.

In recent years, some methods have attempted multi-style editing by disentangling style and linguistic content, where the overall style representation is separated from the content representation to enable controllable recombination [3], [6], [13]. This disentanglement allows users to specify multiple style conditions, such as timbre and emotion, which the model recombines to generate speech with the desired styles, as illustrated in Fig. 1(a). In these approaches, the source speech a is used only to provide the linguistic content c , while all other information in a beyond content is discarded. The target stylistic attributes are then generated entirely from the given style conditions. We refer to these works as the **speech style generation paradigm**, which can be formulated as $G(c, sty_1, \dots, sty_n) \rightarrow y$, where

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Speech samples are provided in <https://ychenn1.github.io/SFM-Adapter/>.

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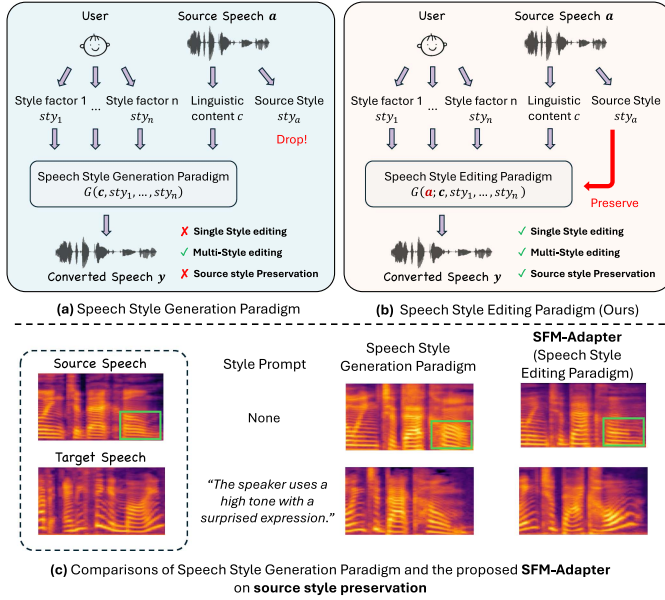


Fig. 1. Motivation. (a) Previous style-generation-based paradigm, (b) style-editing-based paradigm (SFM-Adapter), and (c) evidence that previous methods lack the ability to selectively modify specific aspects of style. As shown in the green box, when the style prompt is absent, the speech style generation paradigm alters the source style, whereas the speech style editing paradigm preserves stylistic characteristics consistent with the source speech.

y is the generated speech. In Fig. 1(c), we compare the impact of the expressive condition when the other components remain the same. As highlighted in the green box, in the absence of an explicit expressive condition, the generated speech exhibits lower expressive similarity to the source speech, rather than naturally preserving the original source style. This suggests that, under such a formulation, it is less straightforward to explicitly modify only selected style attributes while preserving other unspecified source-related attributes.

To better support selective style manipulation, more recent approaches [8], [14] incorporate the full source speech a into the model, enabling interaction between the user-specified style conditions and the existing attributes of the source speech. Formally, this can be viewed as a **speech style editing paradigm**, described as $G(a, c, sty_1, \dots, sty_n) \rightarrow y$, where a provides the complete source speech information and c denotes the linguistic content condition. As illustrated in Fig. 1(b), under this paradigm, the model is designed to modify the style attributes specified by the prompts while preserving the remaining source-related attributes as much as possible. For instance, if the prompt contains only “high pitch”, the model changes pitch while attempting to keep emotion and timbre unchanged; if both timbre and emotion are provided, it edits these prompted attributes jointly.

Under the speech style editing paradigm, we aim to provide flexible and diverse style control over the source speech. To support such fine-grained manipulation, natural-language instructions offer a particularly suitable interface, as they can express a wide range of editing intents beyond predefined style labels. However, this setting requires speech data paired with

detailed style descriptions, which remain scarce in current public datasets. Among the two mainstream model families, audio Large Language Models (LLMs) and diffusion models, prior studies [15], [16] have shown that diffusion-based methods are generally more suitable under limited-data conditions, whereas audio LLM-based models typically require substantially more training data to achieve robust performance. Therefore, in this work, we adopt a reconstruction-based diffusion framework for speech style editing, where the model is trained to reconstruct the original speech from style conditions derived from the same sample.

Building on these ideas, we propose a Style-aware Feature Manipulation Adapter (**SFM-Adapter**) within a Siamese-like diffusion architecture [17] that can incorporate external style conditions and enable precise style manipulation during conversion. As illustrated in Fig. 2, our framework contains a Source Branch that extracts style-and-content features from the input speech, and a Conversion Branch that generates the edited output. We first employ a Multi-Condition Embedder (MCE) to convert two types of style input (a text prompt and an audio prompt) into style features. The text prompt, described in natural language, conveys comprehensive and detailed expressive features. The audio prompt offers precise timbre attributes, compensating for the limitations of natural language descriptions in capturing timbre nuances. The SFM-Adapter employs a learnable style probe, implemented as a set of learnable query vectors, and progressively extracts stylistic information from the source speech through multi-layer cross-attention. During this process, the stylistic features extracted by the MCE interact with the source speech features, facilitating the manipulation of the desired style while maintaining unchanged style attributes. Finally, the manipulated source features and the provided style features are combined as style conditions and integrated into the Conversion Branch through mask attention to complete the style editing. During inference, we apply Classifier-Free Guidance [18] to enhance the controllability of style editing. Additionally, we introduce a Large Audio-Language Model (LALM) that analyzes expressive features and adaptively determines the appropriate speech rate and relative duration changes to further improve editing performance.

The main contributions of this paper are summarized as follows:

- We identify the challenge of inadequate preservation of source speech attributes in the speech style generation paradigm. To address this limitation, we incorporate the source speech and formulate a speech style editing paradigm that supports multiple style editing tasks, enabling controllable manipulation of speech style.
- We propose a Style-aware Feature Manipulation Adapter that explicitly modulates source speech style features conditioned on user-specified style information, and injects the manipulated styles into the editing process via specially designed masked attention.
- We introduce LALM-based length regulation to adaptively adjust the speed of the generated speech, and employ multi-condition sampling to control the strength of style conditions.

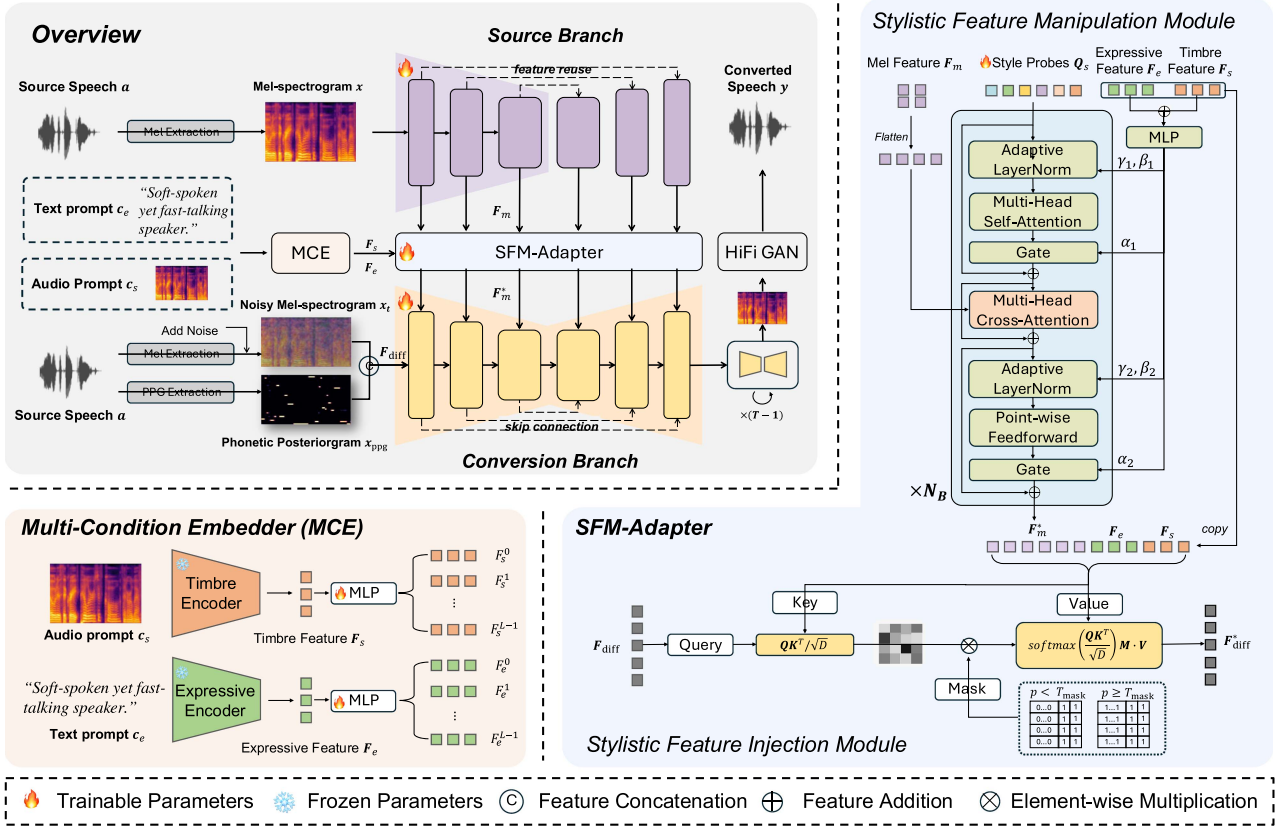


Fig. 2. **Overview of the proposed framework.** The framework is built on a Siamese architecture comprising two branches: the Source Branch for extracting source speech features and a Conversion Branch for reconstructing the edited speech with the guidance of style signals. These two branches are connected through the Multi-Condition Embedder and the SFM-Adapter.

- Experiments show that our framework achieves superior style editing accuracy and produces more natural speech compared to existing methods across multiple SSE tasks.

II. RELATED WORKS

A. Speech Style Editing

Timbre editing: Timbre editing aims to convert the speech of a source speaker to resemble that of a target speaker while preserving linguistic content. A key challenge in this task is to disentangle speaker-specific characteristics from speech without compromising speaker-agnostic representations [7]. Phonetic posteriorgrams (PPGs), which represent frame-level posterior probabilities over phonetic units extracted by an automatic speech recognition model, have been widely adopted for speaker-independent conversion [19], [20]. More recently, self-supervised learning (SSL) methods have been used to enhance content disentanglement in timbre editing models [21], [22], [23], with some leveraging HuBERT [24] and others utilizing WavLM [25].

Expressive style editing: Expressive style editing is a widely studied task that aims to modify the emotional expression of speech while preserving its linguistic content. Some studies use emotion labels [9], [10], [26] to control the expressiveness of generated speech, while others [10], [27] employ averaged

emotion representations derived from emotion classifiers by computing the mean feature vector for a specific emotion category. Additionally, some approaches [8], [11], [12] utilize reference speech as an emotion source, allowing the generated speech to align with the expressive characteristics of the reference.

Multi-style editing: To generate expressive speech, recent studies have explored methods that jointly edit multiple speech attributes, such as timbre, emotion, and speaking speed. Some works adopt a sequence-to-sequence formulation [28], [29] or introduce additional modeling for prosody features [30], [31] to achieve multi-style editing. However, these methods rely on a reference speech to provide the style, which limits their flexibility. For example, if no reference speech with the desired style is available, conversion to that style becomes impossible. Since text can describe speech style from multiple perspectives, and is not constrained by the limitations of using reference speech alone, text prompts have attracted growing attention in recent style editing research. Kuan et al. [32] introduced TextguidedVC, which employs natural language instructions as guidance for the style editing process. PromptVC [33] and HybridVC [6] propose speech-to-speech style editing models that use text prompts to generate style representations for controlling the generated speech. However, these approaches often fail to preserve source speech style attributes that are not explicitly controlled by the text prompts.

B. Denoising Diffusion Probability Models (DDPMs)

Diffusion models [34] have achieved the state-of-the-art sample quality in various tasks, e.g., image generation [35], [36], video generation [37], [38], [39] and audio generation [40], [41], [42]. They model the data distribution by iteratively denoising samples from Gaussian noise, gradually transforming noise into realistic data through a learned reverse diffusion process [43]. Diffusion models have recently been introduced into voice conversion (VC) to model complex speech distributions through iterative denoising processes [43]. Diff-HierVC [30] employs a hierarchical diffusion framework to generate F0 trajectories for expressive zero-shot voice style transfer, achieving improved pronunciation and natural intonation, but its controllability is limited to pitch-related attributes. PromptVC [33] extends diffusion-based style editing by modeling global style representations conditioned on natural language prompts, enabling flexible control over multiple style attributes beyond pitch. In contrast, PromptEVC [44] integrates diffusion models at the emotion embedding level, focusing on fine-grained expressive style editing with explicit control over prosodic patterns and intensity. ClapFM-EVC [45] combines language-audio contrastive pretraining with a conditional flow-matching generative model to enable high-fidelity emotional voice conversion driven by natural language prompts or reference speech.

III. METHOD

A. Problem Definition

Given an input speech sample $\mathbf{a}$ and target styles, the goal is to edit its style to match the target style conditions while preserving all other aspects of $\mathbf{a}$. We use two types of prompts to guide the editing process. For timbre information, we use a reference speech sample $\mathbf{c}_s$ to provide the target speaking timbre characteristics. We also use natural language text descriptions $\mathbf{c}_e$ to provide expressive style attributes such as emotion, speaking rate, and pitch. By using our model G , the generated speech $\mathbf{y}$ can be obtained as:

$$\mathbf{y} = G(\mathbf{a}, \mathbf{c}_e, \mathbf{c}_s). \quad (1)$$

The values $\mathbf{c}_e$ and $\mathbf{c}_s$ can be set to a zero vector $\emptyset$ if needed, indicating that the corresponding condition is not provided, enabling the model to edit only the specified styles while preserving the other aspects of the speech.

B. Overview

As shown in Fig. 2, the proposed framework is built on a Siamese architecture comprising two branches: the Source Branch $S(\cdot)$ and the Conversion Branch $C(\cdot)$. The Conversion Branch adopts a U-Net-based DDPM to generate the edited speech, while the Source Branch, which has the same encoder architecture as the U-Net backbone but with separate model weights, is responsible for extracting multi-scale features from the source speech. These two branches are connected through the Multi-Condition Embedder (MCE) and a Style-aware Feature Manipulation Adapter (SFM-Adapter) (See Section III-D).

During training, the model follows a reconstruction-based DDPM paradigm without paired edited speech supervision. For

each speech sample $\mathbf{a}$, the audio prompt $\mathbf{c}_s$ is randomly cropped from $\mathbf{a}$ to provide timbre guidance, and the text prompt $\mathbf{c}_e$ describes the expressive style of the speech sample. Conditioned on the speech sample and these prompts, the Conversion Branch is trained to predict the Gaussian noise $\epsilon \sim N(0, I)$ added to the mel-spectrogram of $\mathbf{a}$, thereby learning to reconstruct the original mel-spectrogram from its noisy version. During inference, any reference audio prompt and text description can be used to specify the target timbre and expressive style. The audio prompt can be any speech from the target speaker, regardless of its transcript or expressive style. Specifically, the editing process consists of three stages. **First**, the source speech $\mathbf{a}$ is processed to extract a mel-spectrogram $\mathbf{x}$ and PPG features $\mathbf{x}_{\text{ppg}}$, using conventional signal processing methods and a neural PPG model [46], respectively. In the Source Branch, the mel-spectrogram is input to extract multi-scale source speech features $\mathbf{F}_m$. In the Conversion Branch, we add Gaussian noises ϵ to the extracted mel-spectrogram, constructing the noisy mel-spectrogram for training diffusion models. Since PPG features primarily encode linguistic content information [46], they can be treated as content conditions for speech generation. Therefore, we concatenate the PPG features with the noisy mel-spectrogram along the channel dimension and feed them into the Conversion Branch to enhance linguistic accuracy. Simultaneously, both text and audio prompts are processed through the Multi-Condition Embedder, generating stylistic features. **Then** $\mathbf{F}_m$ is interacted with the extracted timbre features $\mathbf{F}_s$ and expressive features $\mathbf{F}_e$ to obtain $\mathbf{F}_m^*$. Subsequently, the noisy features $\mathbf{F}_{\text{diff}}$ integrate the stylistic features from $\mathbf{F}_m^*$, $\mathbf{F}_s$, and $\mathbf{F}_e$ to the conversion diffusion process via the mask-attention. **Finally**, the Conversion Branch outputs the corresponding denoised mel-spectrogram, which is then decoded to a waveform using a HiFi-GAN [47], a high-fidelity neural vocoder widely adopted in previous speech synthesis and voice conversion works due to its high reconstruction quality [7], [12].

C. Multi-Condition Embedder

Given the text prompt $\mathbf{c}_e$ and audio prompt $\mathbf{c}_s$, a Multi-Condition Embedder is employed to extract stylistic features. Specifically, a pretrained timbre encoder $\mathcal{E}_s$ is utilized to extract the timbre features $\mathbf{F}_s = \mathcal{E}_s(\mathbf{c}_s) \in \mathbb{R}^{N_s \times D_s}$, where N_s denotes the number of timbre features, and D_s represents the dimension of the timbre features. Similarly, a pretrained text encoder $\mathcal{E}_e$ is used to extract the expressive features $\mathbf{F}_e = \mathcal{E}_e(\mathbf{c}_e) \in \mathbb{R}^{N_e \times D_e}$, where N_e and D_e denote the number and dimension of the expressive features, respectively.

To provide multi-level stylistic guidance for style manipulation, we employ several projection layers to transform the input style features $\mathbf{F}_e$ and $\mathbf{F}_s$ into a set of features $\{\mathbf{F}_e^l\}_{l=0}^{L-1}$ and $\{\mathbf{F}_s^l\}_{l=0}^{L-1}$, where L is the number of layers in Conversion Branch.

D. Sfm-Adapter

We propose the SFM-Adapter to manipulate the source speech features $\mathbf{F}_m$ using the stylistic features $\mathbf{F}_s$ and $\mathbf{F}_e$. It is also designed to preserve useful source information while mitigating potential source style leakage. The resulting manipulated source

features are then injected into the Conversion Branch. As illustrated in Fig. 2, the SFM-Adapter comprises two modules: the Stylistic Feature Manipulation Module and the Stylistic Feature Injection Module. These modules are integrated into every layer of the Conversion Branch. For simplicity, we omit the layer index in the notation in this section.

Stylistic Feature Manipulation Module: Inspired by Querying Transformer (Q-Former) [48], we randomly initialize learnable style queries $\mathbf{Q}_0 \in \mathbb{R}^{N_q \times D}$ to query out features from the source branch and input style conditions for manipulating stylistic features. N_q is the number of learnable query tokens, and D represents the hidden feature dimension. As shown in Fig. 2, the Stylistic Feature Manipulation Module consists of N_B transformer blocks. Each block contains one multi-head self-attention layer, one multi-head cross-attention layer, one point-wise feedforward layer, and two adaptive layernorm (AdaLN) [49] layers. This design enables controlled feature interaction by selectively extracting style-related information from the source branch through cross-attention, while utilizing AdaLN-based modulation to align the extracted source features with the target style features. Instead of directly reusing the full source representation, the learnable queries encourage the model to focus on source information that is most relevant for style manipulation, which helps reduce potential source style leakage.

The computational process of the stylistic feature manipulation module is as follows:

$$\begin{aligned} (\alpha_1^n, \alpha_2^n, \beta_1^n, \beta_2^n, \gamma_1^n, \gamma_2^n) &= \text{MLP}_n(\mathbf{F}_e, \mathbf{F}_s), \\ \mathbf{Q}_n'' &= \mathbf{Q}_{n-1} + \alpha_1^n \times \text{MSA}_n(\text{AdaLN}(\mathbf{Q}_{n-1}, \gamma_1^n, \beta_1^n)), \\ \mathbf{Q}_n' &= \mathbf{Q}_n'' + \text{MCA}_n(\mathbf{Q}_n'', \mathbf{F}_m), \\ \mathbf{Q}_n &= \mathbf{Q}_n' + \alpha_2^n \times \text{FFN}_n(\text{AdaLN}(\mathbf{Q}_n', \gamma_2^n, \beta_2^n)), \end{aligned} \quad (2)$$

where $n \in [1, 2, \dots, N_B]$ indicates the n -th block, $\mathbf{F}_m$ are the source speech features extracted from the Source Branch, $\text{MSA}_n(\cdot)$ is a multi-head self-attention layer, $\text{MCA}_n(\cdot)$ is a multi-head cross-attention layer, $\text{MLP}_n(\cdot)$ is a multilayer perceptron, $\text{AdaLN}(\cdot)$ is adaptive layer normalization, and $\text{FFN}(\cdot)$ is a feed-forward network. After N_B iterations, we can obtain $\mathbf{Q}_{N_B}$. We denote $\mathbf{Q}_{N_B}$ as $\mathbf{F}_m^*$, since it represents the fused representation of the source style information and the target style information.

Stylistic Feature Injection Module: To integrate the target style into the denoising Conversion Branch, we employ a mask-aware cross-attention mechanism. Given the noise features $\mathbf{F}_{\text{diff}} \in \mathbb{R}^{N \times D}$ as a query, which are obtained by concatenating the noised mel-spectrogram at a diffusion timestep t with the corresponding PPGs, the attention module aggregates stylistic information, guiding the model towards the desired style patterns during the denoising process.

Specifically, we concatenate expressive features $\mathbf{F}_e \in \mathbb{R}^{N_e \times D}$, the timbre features $\mathbf{F}_s \in \mathbb{R}^{N_t \times D}$, and the manipulated source features $\mathbf{F}_m^*$ to strengthen the influence of the target style: $\mathbf{F}_{\text{cond}} = \text{concat}(\mathbf{F}_m^*, \mathbf{F}_e, \mathbf{F}_s)$. By explicitly strengthening the target style features during feature integration, the model is encouraged to rely more on the desired target attributes rather than implicitly preserving the original source style. Then, the

output $\mathbf{F}_{\text{diff}}^*$ of the cross-attention is formulated as follows:

$$\mathbf{F}_{\text{diff}}^* = \text{softmax} \left(\frac{\mathbf{F}_{\text{diff}}(\mathbf{F}_{\text{cond}})^T}{\sqrt{D}} \right) \mathbf{M} \cdot \mathbf{F}_{\text{cond}}, \quad (3)$$

where $\mathbf{M}$ is an attention mask applied to the attention weights to selectively suppress contributions from features derived from the Source Branch during training, defined as follows:

$$\mathbf{M} = \begin{cases} [\mathbf{1}_{L \times N_q} \quad \mathbf{1}_{L \times (N_e + N_s)}], & \text{if } p \geq T_{\text{mask}} \\ [\mathbf{0}_{L \times N_q} \quad \mathbf{1}_{L \times (N_e + N_s)}], & \text{if } p < T_{\text{mask}} \end{cases}, \quad (4)$$

where $\mathbf{1}$ and $\mathbf{0}$ denote the mask with all ones and zeros elements, respectively. The value $p \sim \mathcal{U}(0, 1)$ is a random number uniformly sampled for each sample during the training process, and T_{mask} is a threshold used to control the masking ratio. When p is lower than T_{mask} , the query cannot attend the feature of $\mathbf{F}_m^*$. This random feature discarding helps prevent the model from over-relying on source-derived features and reduces the risk of blindly copying undesired source style cues during training.

E. Training Objectives and Classifier-Free Guidance

During training, we optimize the conditional denoising model $\epsilon_\theta(\text{concat}(\mathbf{x}_t, \mathbf{x}_{\text{ppg}}), t, \mathbf{x}, \mathbf{c}_e, \mathbf{c}_s)$ to predict the added noise using a standard MSE loss:

$$\mathcal{L}_{\text{mse}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\text{concat}(\mathbf{x}_t, \mathbf{x}_{\text{ppg}}), t, \mathbf{x}, \mathbf{c}_e, \mathbf{c}_s)\|^2], \quad (5)$$

where $t \sim [1, T]$ is the denoising timestep that is sampled from the uniform distribution, $\mathbf{x}_t$ is the noisy mel features at timestep t , and ϵ is the ground truth noise. The symbols $\mathbf{x}_{\text{ppg}}$, $\mathbf{x}$, $\mathbf{c}_e$ and $\mathbf{c}_s$ denote PPG feature, source mel-spectrogram, text condition and audio condition, respectively.

To improve DDPM with fewer sampling steps, we employ an additional loss term $\mathcal{L}_{\text{vlb}}$, corresponding to the variational lower bound, to learn the model variance Σ_θ , following [50].

$$\mathcal{L}_{\text{vlb}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \text{concat}(\mathbf{x}_t, \mathbf{x}_{\text{ppg}}), t, \mathbf{x}, \mathbf{c}_e, \mathbf{c}_s))], \quad (6)$$

where $D_{\text{KL}}(\cdot \| \cdot)$ denotes the Kullback–Leibler (KL) divergence, which measures the discrepancy between the true posterior distribution $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ and the model-predicted distribution $p_\theta(\mathbf{x}_{t-1} | \cdot)$.

The overall training objective of the proposed framework is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{vlb}}. \quad (7)$$

To enable controllable generation, we further adopt Classifier-Free Guidance (CFG) [18], [41], [42], [50]. CFG is a guidance strategy for diffusion models that improves conditional generation by jointly training conditional and unconditional denoising models, without relying on an external classifier. During training, the conditional inputs $\mathbf{c}_e$ and $\mathbf{c}_s$ are randomly dropped with a fixed probability (e.g., 10%) in (5), allowing the model to learn both conditional and unconditional denoising behaviors.

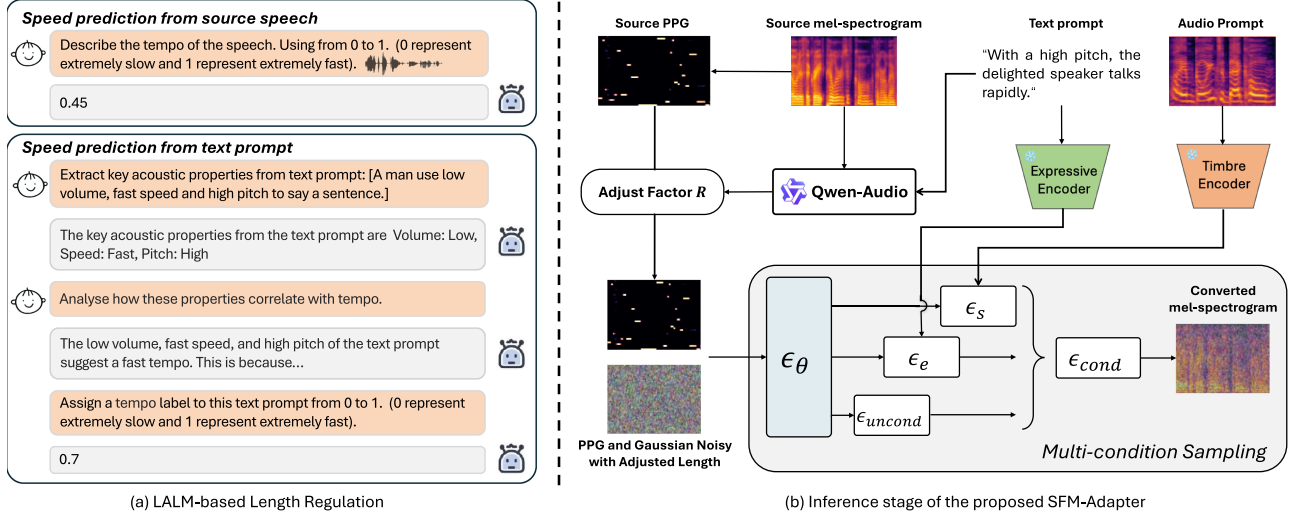


Fig. 3. Illustration of LALM-based Length Regulation and the inference process of SFM-Adapter. In the inference stage of SFM-Adapter, we adopt an adaptive sampling strategy, including LALM-based Length Regulation and Multi-condition Sampling.

F. Adaptive Sampling

1) *LALM-Based Length Regulation*: The speed and style of speech are correlated, as style can influence the speed of speech [1]. Given that LALMs exhibit strong comprehension of both text and audio [51], we propose LALM-based Length Regulation, which leverages LALMs' understanding of style to predict the target speaking speed and adjust the duration of the speech accordingly during style editing. In our work, we use Qwen-Audio [51] as it demonstrates strong capabilities in speech understanding.

To modulate the length based on the source speech, we first extract its speed. As shown in Fig. 3(a), we design a prompt to obtain the speed of the source speech, resulting in a speed $r_{\text{src}} = \text{LALM}(x)$, where $\text{LALM}(\cdot)$ denotes the large audio-language model. To extract the target speed from text prompts, we employ a chain-of-thought (CoT) approach [52]. As illustrated in Fig. 3(a), the extraction task is decomposed into multiple subtasks, allowing us to progressively obtain the target speed, represented as $r_{\text{trg}} = \text{LALM}(c_e)$. We adopt a logarithmic speed adjustment approach, which better aligns with human sensitivity to speed changes compared to linear adjustment, particularly in transitions between fast and slow speeds [53]. Thus, the speed adjustment factor is defined as $R = \left(\frac{r_{\text{trg}}}{r_{\text{src}}}\right)^\eta$, where $\eta \in (0, 1)$ is a smoothing coefficient used to reduce the sensitivity to extreme rate differences. In our experiments, we set $\eta = \frac{1}{\ln 10}$.

By constraining extreme speed changes, the formulation is designed to produce speed transformations that are more natural and better aligned with human perception during speed editing. As illustrated in Fig. 3, the factor R is applied to PPG features. We perform speed interpolation by scaling the length of the PPG features according to R . Given that L_{src} represents the original length of the PPG features and L_{trg} denotes the desired length, the interpolated length is computed as: $L_{\text{trg}} = \frac{L_{\text{src}}}{R}$. The randomly initialized noise x_T is also manipulated in length using

R to align with the PPG features, resulting in adjusted Gaussian noise x'_T . The adjusted noise is then concatenated with the PPG features along the channel dimension and used as the input for network inference, *i.e.*, $\text{concat}(x'_T, x_{\text{ppg}})$. In this setting, if r_{trg} is greater than r_{src} , the resulting L_{trg} will be shorter than L_{src} , reflecting a faster speech speed. This scaling adjusts the temporal length of the PPG features to match the target speed, enabling the generated speech to follow the speed specified by the text prompt c_e .

2) *Multi-Condition Sampling*: Once the model learns style-based speech editing, inference is performed by first sampling a Gaussian noise x_T and then denoising x_T in an iterative manner using the DDPM. To flexibly control the influence of conditions F_s and F_e , we adopt classifier-free guidance [18] in our multi-conditional sampling process. The procedure is illustrated in Fig. 3(b). The denoising procedure is defined as follows:

$$\epsilon_{\text{cond}} = \epsilon_{\text{uncond}} + w_e \epsilon_e + w_s \epsilon_s \quad (8)$$

where $\epsilon_{\text{uncond}} = \epsilon_\theta(\text{concat}(x'_t, x_{\text{ppg}}), t, x, \emptyset, \emptyset)$ denotes the model's unconditioned prediction, and x'_t is the noisy mel-spectrogram at noising step t . Here, both the text and audio prompt conditions are set to zero. The text-guided prediction and the speech-guided prediction are represented by $\epsilon_e = \epsilon_\theta(\text{concat}(x'_t, x_{\text{ppg}}), t, x, c_e, \emptyset) - \epsilon_{\text{uncond}}$ and $\epsilon_s = \epsilon_\theta(\text{concat}(x'_t, x_{\text{ppg}}), t, x, \emptyset, c_s) - \epsilon_{\text{uncond}}$, respectively. The guidance scales corresponding to the text prompt and audio prompt are denoted as w_e and w_s , respectively. A higher w_e enhances style expressiveness, while a larger w_s strengthens speaker timbre preservation.

IV. EXPERIMENTS

A. Datasets

We conduct our experiments on PromptSpeech [2], Emotional Speech Dataset (ESD) [54], and VCTK [55]. (1) **PromptSpeech** contains over 26,000 real speech samples from LibriTTS [56],

each paired with a corresponding text style prompt. The text style prompt in PromptSpeech covers four style factors: gender, pitch, speaking speed, and loudness. (2) **ESD** includes speech samples from 10 native English speakers and 10 native Chinese speakers, covering five emotion categories: neutral, happy, angry, sad, and surprised. ESD contains 350 distinct utterances in total, where 300 are for training, 20 for validation, and 30 for testing. TextrolSpeech [13] extends ESD by providing text style prompts for its speech samples, incorporating not only the four style factors from PromptSpeech but also an emotion factor. (3) **VCTK** dataset comprises speech samples from 107 speakers. During training, we select 19,400 speech samples from PromptSpeech and 16,500 from ESD, yielding a total of 35,900 training samples. For multi-style editing evaluation, we construct 1,000 speech pairs from ESD. Each pair consists of a source and target speech with the same linguistic content. The expressive style editing evaluation set follows a setup similar to the multi-style setting, with the only difference being that the source and target speech originate from the same speaker. Both evaluation settings use speech samples whose linguistic content is unseen during training. For timbre editing evaluation, we randomly constructed 1,000 speech pairs from VCTK, each comprising a source and a target speech with the same linguistic content but different timbre.

B. Implementation Details

In our experiments, all speech clips are sampled at 16 kHz. The mel-spectrograms are computed from the raw waveforms using a frame size of 1024, a hop size of 256, and 80 mel frequency channels. We employ the HiFi-GAN vocoder [47] to reconstruct waveforms from the generated mel-spectrograms. To extract text prompt features, we utilize a pretrained AngIE model [57], producing 1024-dimensional embeddings. For speaker prompt features, we use a pretrained ECAPA-TDNN model [58], trained on the VoxCeleb2 dataset [59], to extract a 192-dimensional global timbre embedding representing the target speaker.

We train our network on two NVIDIA A100 GPUs for 30,000 iterations with a total batch size of 64, and the AdamW optimizer [60] is used with a learning rate of 0.0001. For a trade-off between computational efficiency and editing quality, we set the number of Q_s to 8 and use $N_B = 2$ blocks in the stylistic feature manipulation module. The masking threshold T_{mask} is fixed at 0.3. During inference, mel-spectrograms are generated using $T = 50$ denoising steps, with guidance scales set to $w_e = 2$ for the text prompt and $w_s = 2$ for the audio prompt. The LALM-based length regulation adopts deterministic decoding. The exact prompts are provided in our demo page.

C. Compared Methods

To comprehensively evaluate our method under different editing scenarios, we conduct three groups of experiments: (1) multi-style editing, (2) expressive style editing, and (3) timbre editing. Since these tasks differ in their objectives and assumptions, we select task-specific baseline methods for fair comparison.

In particular, Vevo [8] is a unified framework that supports multiple editing functionalities, including three distinct settings: joint timbre-expressive style editing, expressive style editing only, and timbre editing only. Following the original paper, we denote these variants as Vevo-Voice, Vevo-Style, and Vevo-Timbre, respectively, and use them in the corresponding experimental settings.

For multi-style editing, we compare our method with Vevo-Voice, which jointly performs timbre and expressive style editing. For expressive style editing, we compare against StyleVC [61], AINN [12], and Vevo-Style, all of which are designed for expressive style editing and trained on the ESD dataset. For timbre editing, we include StyleVC [61], DDDM-VC [62], Diff-HierVC [30], Vevo-Timbre, and FreeVC [23], as these methods specifically focus on speaker timbre editing. All baselines are trained and evaluated on the same datasets as our method to ensure fair comparison.

D. Evaluation Metrics

We perform both objective evaluation and human subjective evaluation. The objective metrics include Word Error Rate (**WER**), Speaker Embedding Cosine Similarity (**SECS**), Pitch Pearson Correlation (**Corr**) [3], Emotion Classification Accuracy (**ACC**) and Unified Speech Mean Opinion Score Predictor (**UTMOS**) [3]. Specifically, WER is calculated based on Whisper-large-v3 [63] to evaluate the generated speech's intelligibility. SECS measures the speaker similarity between the generated speech and the audio prompt, which is calculated using the cosine similarity between speaker embeddings extracted from the ECAPA-TDNN model [58]. To evaluate the quality of the generated speech, we use UTMOS to predict the Mean Opinion Score (MOS). We additionally adopt the Corr and ACC metrics to measure the performance of the style editing. Corr is employed to measure the correlation between the pitch contours of generated speech and target speech. It evaluates how well the pitch of the generated speech aligns with that of the target speech. Additionally, we use a pretrained emotion classifier ACRNN [64] to evaluate the accuracy of the generated speech (ACC). For subjective evaluation, we employ the naturalness Mean Opinion Score (**nMOS**) to assess the naturalness of the generated samples. We further use two similarity MOS metrics: **sMOS-e**, which evaluates the expressive similarity between the generated speech and the text style prompt, and **sMOS-s**, which evaluates the timbre similarity between the generated speech and the reference audio prompt. To evaluate cross-attribute preservation, we define additional metrics for non-target attribute preservation. Specifically, in expressive style editing, $SECS_{src}$ is used to measure whether the source speaker identity is preserved after modifying expressive style attributes. In timbre editing, $Corr_{src}$ and ACC_{src} are used to measure whether the original expressive style is preserved after editing speaker timbre.

Human Evaluation: We conduct a human subjective evaluation to assess the quality of generated speech across three tasks: multi-style editing, expressive style editing, and zero-shot timbre editing. The evaluation set used for the listening test consists of 60 groups, randomly sampled from the full evaluation set,

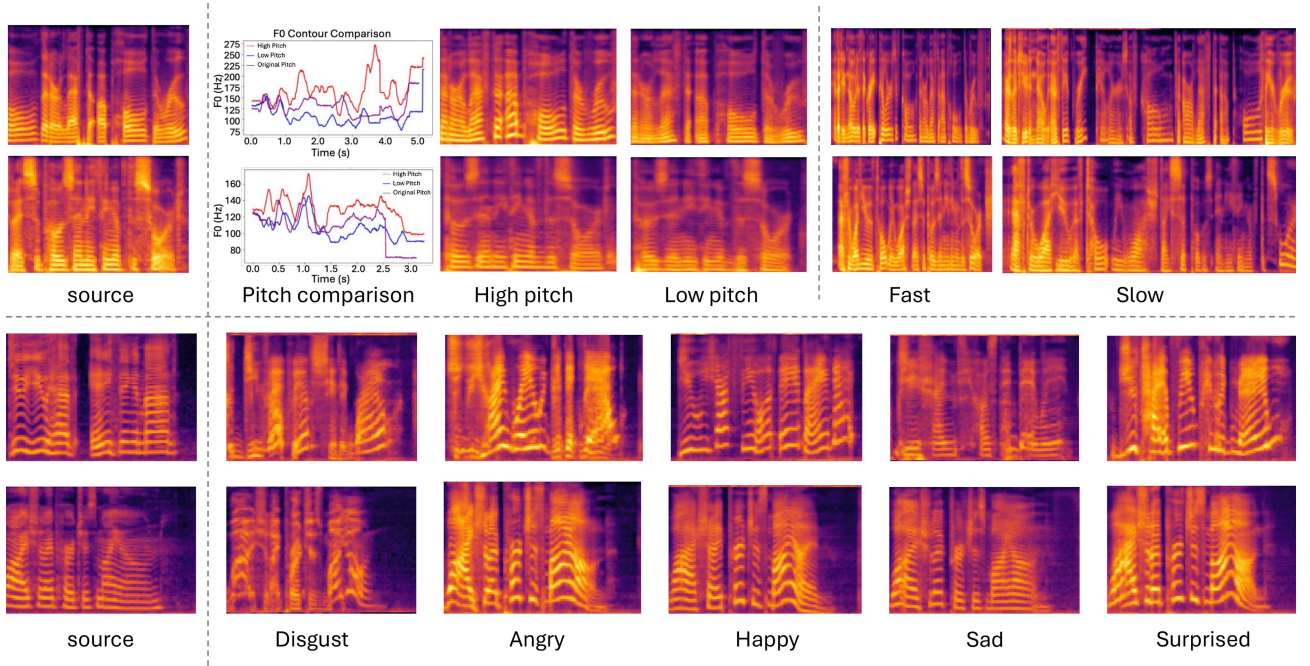


Fig. 4. Visualization of speech style editing. The upper section illustrates pitch and speed editing. In the pitch comparison, the red line represents high pitch, the purple line corresponds to the pitch of the source speech, and the blue line represents low pitch. The lower section presents expressive style editing across five emotion categories.

TABLE I

THE SUBJECTIVE AND OBJECTIVE EVALUATION RESULTS FOR OUR METHODS AND THE BASELINE SYSTEMS IN **MULTI-STYLE EDITING**. ALL SUBJECTIVE METRICS ARE COMPUTED WITH 95 % CONFIDENCE INTERVALS AND “GT” REFERS TO GROUND TRUTH SAMPLES.

Methods	WER ↓	SECS ↑	UTMOS ↑	Corr ↑	ACC ↑	nMOS ↑	sMOS-s ↑	sMOS-e ↑
GT	1.77	-	3.82	-	0.98	4.01 ± 0.12	-	-
Vevo-Voice [8]	4.91	0.61	3.60	0.17	0.53	3.74 ± 0.23	3.64 ± 0.26	3.38 ± 0.23
SFM-Adapter (ours)	6.17	0.64	3.73	0.38	0.70	3.78 ± 0.26	3.70 ± 0.29	3.70 ± 0.30

with 20 groups for each task. Naturalness (nMOS), emotional similarity (sMOS-e), and timbre similarity (sMOS-s) scores are assigned by 20 participants who are proficient in English.

E. Performance

We conduct performance comparisons on both multi- and single-style editing settings to demonstrate the editing ability of our framework. Fig. 4 presents visualizations of the generated speech under single-style editing settings. For pitch editing, speech generated with the “high pitch” text prompt exhibits clearly elevated pitch contours compared to that generated with the “low pitch” prompt. Similarly, for expressive style editing, the mel-spectrograms show distinct acoustic differences: the “Sad” style is characterized by flatter pitch contours and lower energy, whereas the “Surprised” style displays rising pitch and higher energy. These observations demonstrate that the proposed model can manipulate the intended acoustic features in response to different style prompts.

Performance on multi-style editing: We conduct multi-style editing experiments to evaluate the effectiveness of our model in handling diverse style editing tasks. We use the caption of

the target speech as the text prompt to provide expressive style, and use the target speech as the audio prompt to provide timbre style. As shown in Table I, ¹ compared with the state-of-the-art Vevo-Voice [8], SFM-Adapter achieves a higher SECS score of 0.64, along with a Corr of 0.38 and an ACC of 0.70, showing an improvement over Vevo-Voice. SFM-Adapter achieves an sMOS-e of 3.70 and an sMOS-s of 3.70, compared to 3.64 and 3.38 achieved by Vevo-Voice, respectively. This indicates that the generated speech from SFM-Adapter is perceived as more similar to the target in both expressive style and timbre style. Meanwhile, the naturalness of the generated speech is well preserved, with an nMOS of 3.78. Compared to Vevo-Voice, these results highlight the effectiveness of our proposed method. This result also indicates that diffusion-based editing remains competitive and practical under limited paired-data conditions, even when compared with large-scale LLM-based approaches.

Performance on expressive style editing: As shown in Table II, our proposed SFM-Adapter outperforms the previous methods

¹TextguidedVC [32], HybridVC [6] and PromptVC [33], did not release implementation code and training datasets, so we cannot use them for direct comparison.

TABLE II

THE SUBJECTIVE AND OBJECTIVE EVALUATION RESULTS FOR OUR METHODS AND THE BASELINE SYSTEMS IN **EXPRESSIVE STYLE EDITING**. “CONDITION” REFERS TO THE TYPE OF USER-PROVIDED INPUT USED DURING EDITING. ALL SUBJECTIVE METRICS ARE COMPUTED WITH 95 % CONFIDENCE INTERVALS AND “GT” REFERS TO GROUND TRUTH SAMPLES.

Methods	Condition	WER ↓	Corr ↑	ACC ↑	SECS _{src} ↑	UTMOS ↑	nMOS ↑	sMOS-e ↑
GT	-	1.77	-	0.99	-	3.97	4.12 ± 0.13	-
StyleVC [61]	audio	10.80	0.10	0.23	0.64	3.69	2.97 ± 0.38	2.51 ± 0.41
AINN [12]	audio	13.67	0.21	0.50	0.62	2.37	2.92 ± 0.29	2.41 ± 0.36
Vevo-Style [8]	audio	3.57	0.24	0.50	0.70	3.60	3.49 ± 0.27	3.37 ± 0.33
SFM-Adapter (ours)	text	5.30	0.35	0.73	0.72	3.74	3.95 ± 0.28	4.09 ± 0.27

TABLE III

THE SUBJECTIVE AND OBJECTIVE EVALUATION RESULTS FOR SFM-ADAPTER AND THE BASELINE SYSTEMS IN ZERO-SHOT **TIMBRE EDITING**. ALL SUBJECTIVE METRICS ARE COMPUTED WITH 95 % CONFIDENCE INTERVALS AND “GT” REFERS TO GROUND TRUTH SAMPLES. * INDICATES FREEVC IS NOT IN A ZERO-SHOT SETTING.

Methods	WER ↓	SECS ↑	Corr _{src} ↑	ACC _{src} ↑	UTMOS ↑	nMOS ↑	sMOS-s ↑
GT	3.21	-	-	-	4.01	4.03 ± 0.06	-
StyleVC [61]	25.66	0.41	0.34	0.56	3.38	2.90 ± 0.38	3.02 ± 0.37
DDDM-VC [62]	9.35	0.39	0.39	0.74	3.46	3.04 ± 0.33	3.02 ± 0.32
Diff-HierVC [30]	8.40	0.40	0.41	0.78	3.67	3.02 ± 0.28	2.99 ± 0.35
Vevo-Timbre [8]	8.25	0.61	0.43	0.79	3.85	3.81 ± 0.34	3.67 ± 0.30
FreeVC* [23]	8.04	0.47	0.37	0.76	3.84	3.43 ± 0.23	3.28 ± 0.34
SFM-Adapter (ours)	7.95	0.66	0.46	0.80	3.88	3.72 ± 0.27	3.69 ± 0.32

across all three style-related metrics. In terms of style, the previous best method, Vevo-style, achieves an ACC score of 0.50, while our proposed SFM-Adapter method surpasses it with a higher ACC score of 0.73. SFM-Adapter also attains the best Corr score of 0.35, improving over Vevo-Style by 0.11. In addition, SFM-Adapter achieves the highest SECS_{src}, indicating that it can effectively preserve the source speaker identity while modifying the target style attribute. In terms of speech quality, SFM-Adapter achieves the highest UTMOS of 3.74 and an nMOS of 3.95, compared to 3.97 and 4.12 for real speech, respectively. These results suggest that our method generates speech with a relatively high degree of naturalness, while still falling slightly short of real human speech. The subjective evaluation results on emotion similarity are mostly consistent with the objective metrics, which achieve an sMOS-e of 4.09, confirming the effectiveness of SFM-Adapter.

Performance on timbre editing: We evaluate our timbre editing performance against several methods in a zero-shot setting, where both the source and target speakers are unseen during training. As shown in Table III, SFM-Adapter achieves a SECS score of 0.66 for speaker similarity, while most baseline methods fall below 0.5, reflecting a noticeable gap in performance. A similar trend is observed in the sMOS-s scores, further indicating that our method effectively enhances resemblance to the target speaker. Additionally, our method achieves a competitive WER compared to baseline methods, suggesting that it maintains strong intelligibility in timbre editing. In terms of naturalness, SFM-Adapter surpasses the baseline methods, reaching a UTMOS score of 3.88, merely 0.13 below the ground truth, indicating that our method can generate speech with reasonable naturalness. We further observe that SFM-Adapter achieves the highest Corr_{src} and a competitive ACC_{src}, indicating that it better preserves the original expressive style of the source speech during timbre editing than compared methods.

TABLE IV

ABLATION STUDY ON MODEL DESIGN. THE BASELINE IS THE MODEL WITHOUT USING SOURCE BRANCH FOR SOURCE MEL-SPECTROGRAM, AND ONLY USING CROSS-ATTENTION MECHANISM FOR FEATURE FUSION.

Model	WER ↓	SECS ↑	UTMOS ↑	Corr ↑	ACC ↑
baseline	8.32	0.58	3.72	0.33	0.55
SFM-Adapter (Full)	6.17	0.64	3.73	0.38	0.70
w/o Modulation	7.63	0.61	3.72	0.35	0.68
w/o Mask	5.12	0.42	3.73	0.26	0.53
w/o Cat style Fea.	7.68	0.47	3.74	0.31	0.62
w/o CFG	6.35	0.61	3.72	0.34	0.64
w/o Length Regulation	6.26	0.63	3.71	0.32	0.66

Compared with Vevo, SFM-Adapter shows a different trade-off between intelligibility and style controllability. Although SFM-Adapter yields slightly higher WER than Vevo in Tables I and II, this gap is likely related to the increased difficulty of language-guided style editing, where the model must simultaneously preserve linguistic accuracy and follow flexible style descriptions. This interpretation is further supported by the timbre editing results in Table III: without natural-language style control, SFM-Adapter achieves lower WER than Vevo-Timbre (7.95 vs. 8.25). Moreover, Vevo is an LLM-based method trained on approximately 60,000 hours of speech, whereas SFM-Adapter uses only 60 hours of training data. Despite this large difference in training scale, SFM-Adapter still achieves comparable WER while obtaining stronger style-related performance.

F. Ablation Studies and Analysis

Effect of model design: We conduct an ablation study under the multi-style editing setting to prove the effectiveness of the proposed designs. The baseline method directly injects style features using naive cross-attention. As shown in Table IV, we observe a clear performance drop across all metrics for the baseline method, suggesting that relying solely on the cross-attention

TABLE V
ABLATION STUDY OF MASKING RATIO ACROSS DIFFERENT TASKS. SECS/CORR/ACC METRICS ARE TASK-SPECIFIC.

Masking Ratio	Multi-style editing				Expressive style editing				Timbre style editing				Normalized Average
	WER ↓	SECS ↑	Corr ↑	ACC ↑	WER ↓	SECS _{src} ↑	Corr ↑	ACC ↑	WER ↓	SECS ↑	Corr _{src} ↑	ACC _{src} ↑	
0%	5.12	0.42	0.26	0.53	4.23	0.80	0.24	0.43	5.81	0.49	0.57	0.87	0.50
15%	5.69	0.51	0.32	0.64	4.78	0.77	0.30	0.63	7.09	0.55	0.52	0.81	0.57
30%	6.17	0.64	0.38	0.70	5.30	0.72	0.35	0.73	7.95	0.66	0.46	0.80	0.65
45%	6.19	0.67	0.39	0.71	5.51	0.64	0.36	0.74	8.32	0.65	0.40	0.70	0.57
60%	6.23	0.68	0.39	0.72	5.62	0.48	0.35	0.74	8.39	0.68	0.36	0.64	0.49

mechanism may be insufficient for effective style editing. In the design of Stylistic Feature Manipulation Module, we compare the results “w/o Modulation” by removing the AdaLN, it can be observed that the style expression in the generated speech is weakened. This is evident from the reductions in SECS, Corr, and ACC compared to SFM-Adapter, highlighting that modulation plays an important role in improving style editing. Thus, combining the manipulated features with style features is the most effective approach to guide the network in style editing, allowing the generated speech to closely reflect the intended styles. In the design of the Stylistic Feature Injection Module, we ablate the effect of mask attention and the integration of style features. Compared to SFM-Adapter, removing the input style feature (“w/o Cat style Fea.”) results in a decrease in style-related metrics, with SECS decreasing by 0.17, ACC by 0.08, and Corr by 0.07. This demonstrates that the provided style features can further reinforce the performance of style editing. Additionally, when the mask in the SFM-Adapter block is removed (“w/o Mask”), the WER improves, decreasing from 6.17 to 5.12. However, SECS, ACC, and Corr show noticeable declines, suggesting that without the mask, the network relies more heavily on the input mel-spectrogram during training and pays less attention to the style information provided by the input conditions. This highlights the effectiveness of the mask in guiding the network to better utilize the provided style conditions for editing. We also evaluate the impact of classifier-free guidance (CFG) and Length Regulation. Removing CFG slightly reduces SECS, Corr, and ACC, indicating it strengthens conditional guidance during sampling. Removing Length Regulation mainly degrades expressive style metrics (Corr and ACC), confirming its role in realizing the target expressive style.

Effect on mask ratio: In our SFM-Adapter, the masking ratio controls the proportion of source features dropped before being incorporated into the conversion branch. To examine its impact, we conduct an ablation study with masking ratios of 0%, 15%, 30%, 45%, and 60%. To provide an overall comparison across different masking ratios, we compute an overall score by first normalizing each metric independently across all masking-ratio settings and then averaging the normalized values over all metrics. The results in Table V show a clear trade-off controlled by the masking ratio. Increasing the masking ratio generally improves style editing performance, as reflected by higher SECS/Corr/ACC scores, but also leads to higher WER and weaker preservation of some source-related attributes. In particular, for expressive style editing, excessive masking causes a notable drop in SECS_{src}, indicating reduced preservation of source speaker information. Overall, 30% masking gives the best

TABLE VI
ROBUSTNESS ANALYSIS UNDER EXTREME SOURCE-TARGET STYLE CONFLICTS
“F” INDICATES FEMALE AND “M” INDICATES MALE

Source → Target	SECS ↑	ACC ↑	Corr ↑
happy, high-pitch F → sad, low-pitch M	0.61	0.66	0.35
sad, low-pitch M → surprise, high-pitch F	0.65	0.69	0.37
angry, high-pitch M → sad, low-pitch F	0.63	0.68	0.34

TABLE VII
SOURCE SPEECH ATTRIBUTES PRESERVATION IN EXPRESSIVE STYLE EDITING

Methods	ACC ↑	Corr ↑	SECS _{src} ↑
baseline	0.71	0.32	0.18
SFM-Adapter	0.73	0.35	0.72

normalized average score, suggesting the best balance between target style control and source information preservation.

Robustness to source-style leakage: To evaluate robustness under source-target style conflicts, we construct three challenging editing settings with explicitly conflicting source and target attributes, including emotion, pitch, and gender/timbre. For each setting, we randomly sample 25 source-target pairs, where each pair consists of a source speech sample and a target style specification with strongly mismatched attributes. Table VI shows that the performance of SFM-Adapter on these extreme source-target conflict cases remains close to its average performance across all multi-style editing cases shown in Table I. Averaged over the three conflict cases, the model achieves a SECS of 0.63, an ACC of 0.68, and a Corr of 0.35, compared with 0.64, 0.70, and 0.38 in Table I, respectively. These results indicate that SFM-Adapter remains robust even under highly conflicting editing conditions.

G. Disentanglement Analysis

To further validate the disentanglement capability of the proposed method, we conduct both quantitative analysis and feature-level visualization.

Quantitative analysis: For each editing task, we modify only the specified attribute, while leaving the remaining attributes unspecified. We evaluate (1) editing success by comparing the edited attribute with the *target style*, and (2) attribute preservation by comparing the non-edited attributes with the *source style*. Expressive style editing is evaluated by target-style similarity and source-speaker preservation, while timbre editing is evaluated by target-speaker similarity and source-style preservation. As shown in Tables VII and VIII, our method consistently achieves better cross-attribute preservation compared to baseline

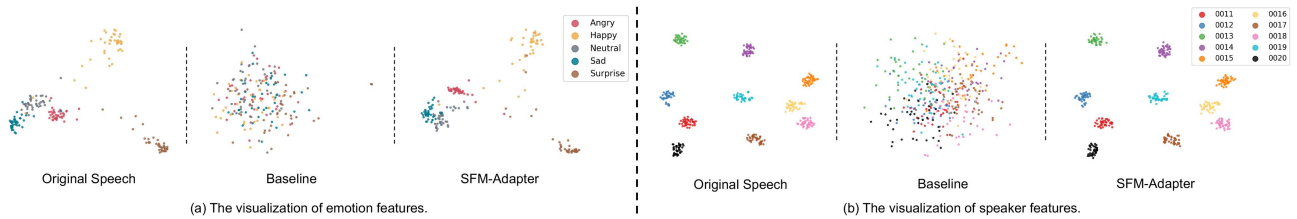


Fig. 5. Visualization of emotion and speaker features under pitch editing. Sub-figure (a) shows the clustering of emotion features, and sub-figure (b) shows the clustering of speaker features for the original speech, the baseline model, and SFM-Adapter. Compared with the baseline, SFM-Adapter preserves clearer cluster structures that remain closer to those of the original speech.

TABLE VIII
SOURCE SPEECH ATTRIBUTES PRESERVATION IN TIMBRE EDITING

Methods	SECS $\uparrow$	CORR _{src} $\uparrow$	ACC _{src} $\uparrow$
baseline	0.63	0.12	0.21
SFM-Adapter	0.66	0.46	0.80

TABLE IX
COMPARISON OF MODEL SIZE AND INFERENCE TIME AMONG DIFFERENT METHODS

Methods	Parameters (M)	Inference time (s)
Diff-HierVC [30]	18.28	1.05
FreeVC [23]	40.80	0.31
DDDM-VC [62]	79.41	0.73
StyleVC [61]	347.88	0.38
AINN [12]	39.83	0.35
Vevo [8]	819.18	2.56
SFM-Adapter (Ours)	226.46 (Adapter: 57.69)	1.54

method. Specifically, during expressive style editing, our model maintains higher speaker similarity, while during timbre editing, it better preserves expressive attributes such as emotion. These results demonstrate that the proposed method can selectively modify the intended attribute while keeping unrelated attributes unchanged, whereas baseline method tend to entangle multiple attributes during editing.

Feature visualization: We further visualize the distributions of emotion and speaker features during pitch editing. As shown in Fig. 5, the features produced by baseline methods become more dispersed and exhibit mixed cluster structures after editing, indicating that non-target attributes are also affected. In contrast, our method preserves clearer cluster boundaries and maintains distributions closer to the original speech. This indicates that the proposed method can effectively modify pitch while preserving emotion and speaker identity, providing further evidence of disentangled representation.

H. Efficiency Analysis

We measure model size by the total number of parameters and evaluate computational complexity in terms of inference time. As shown in Table IX, The proposed SFM-Adapter has 226.46 M parameters in total, of which 57.69 M come from the adapter module. This is substantially smaller than large-scale models like Vevo (819.18 M) and also smaller than StyleVC (347.88 M). In terms of inference efficiency, SFM-Adapter

requires only 1.54 s per speech, outperforming Vevo, which takes 2.56 s per speech. SFM-Adapter consistently achieves superior style editing performance and maintains comparable speech intelligibility to baseline methods. This demonstrates that the added architectural complexity is justified, as it leads to better control and flexibility in multi-conditional style editing without sacrificing efficiency.

I. Evaluation of LALM-Based Length Regulation

We conducted an ablation study to assess the impact of the LALM-based length regulation module on style editing performance. When removing this component, we observed a degradation in ACC. Without length regulation, the ACC has decreased from 0.70 to 0.66, demonstrating that speed alignment plays an important role in perceived expressiveness. Regarding robustness to phrasing variations, we evaluated the LALM's speed estimation results using text prompts, with some prompts containing explicit speed cues and others lacking them. For example, the text prompt “A stunned female voice, fast and high-pitched.” yields a predicted speed of 0.9, while “Her voice was soft and her sad speech unfolded gradually.” results in a speed of 0.2. This demonstrates that LALM can distinguish between prompts that imply speaking rate and those that do not. In addition, the overall accuracy of text-based speed prediction is 0.82, further confirming the reliability of LALM in interpreting speed from natural language descriptions

V. CONCLUSION

In this paper, we have presented SFM-Adapter, a novel framework that supports multiple style editing tasks, enabling controllable manipulation of speech style. Our framework consists of a Source Branch for extracting source speech features and a Conversion Branch for generating speech. We introduce two modules to extract rich style descriptions and integrate stylistic features into the conversion process, respectively. During inference, Adaptive Sampling enhances editing controllability, while a Large Audio-Language Model (LALM) based Length Regulation refines speech rate and duration. Experimental results show that SFM-Adapter achieves more natural and accurate style editing compared to existing methods, while also enabling flexible control across multi-styles. In future work, we plan to explore finer-grained modeling of prosodic attributes, such as intonation patterns, rhythm, and local speaking rate variations, to further enhance style controllability.

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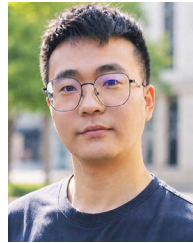


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